Generalized Forces (Part B)

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1. D'Alembert's Principle

Consider a system of N particles, with particle j having mass m_j and position vector $\vec{r}_j$. Newton's second law for each particle reads

$$\vec{F}_j = m_j \ddot{\vec{r}}_j. \tag{1}$$

The total force on particle j is the sum of an applied force $\vec{F}_j^{(a)}$ and a constraint force $\vec{f}_j$, so that

$$\vec{F}_j^{(a)} + \vec{f}_j - m_j \ddot{\vec{r}}_j = \vec{0}. \tag{2}$$

Taking the dot product with a virtual displacement $\delta\vec{r}_j$ and summing over all particles gives

$$\sum_{j=1}^{N} \left(\vec{F}_j^{(a)} + \vec{f}_j - m_j \ddot{\vec{r}}_j \right) \cdot \delta\vec{r}_j = 0. \tag{3}$$

The key assumption is that the constraints are **ideal**: the total virtual work done by the constraint forces vanishes,

$$\sum_{j=1}^{N} \vec{f}_j \cdot \delta\vec{r}_j = 0. \tag{4}$$

The constraint forces then drop out, and we are left with **d'Alembert's principle (达朗贝尔原理)**:

$$\boxed{\sum_{j=1}^{N}\left(\vec{F}_j^{(a)} - m_j\ddot{\vec{r}}_j\right)\cdot\delta\vec{r}_j = 0} \tag{5}$$

for every **admissible virtual displacement**. In words, the virtual work done by the applied forces plus the virtual work done by the inertial forces $-m_j\ddot{\vec{r}}_j$ must vanish. The principle converts a problem in dynamics into a statement about virtual work and removes the constraint forces from the outset.

1.1 Generalized Coordinates and Generalized Forces

Suppose the configuration is described by **generalized coordinates (广义坐标)** $q_1, \dots, q_n$, so that

$$\vec{r}_j = \vec{r}_j(q_1, \dots, q_n, t). \tag{6}$$

A virtual displacement is taken at fixed time, hence

$$\delta\vec{r}_j = \sum_{i=1}^{n}\frac{\partial\vec{r}_j}{\partial q_i}\,\delta q_i. \tag{7}$$

Substituting this into d'Alembert's principle and exchanging the order of summation gives

$$\sum_{i=1}^{n}\left[\sum_{j=1}^{N}\vec{F}_j^{(a)}\cdot\frac{\partial\vec{r}_j}{\partial q_i} - \sum_{j=1}^{N}m_j\ddot{\vec{r}}_j\cdot\frac{\partial\vec{r}_j}{\partial q_i}\right]\delta q_i = 0. \tag{8}$$

The first bracket defines the **applied generalized force (广义力)** associated with q_i:

$$Q_i = \sum_{j=1}^{N}\vec{F}_j^{(a)}\cdot\frac{\partial\vec{r}_j}{\partial q_i}. \tag{9}$$

Here Q_i is the applied generalized force: it is built only from the applied forces $\vec{F}_j^{(a)}$ and receives no contribution from the constraint forces $\vec{f}_j$.

For the second bracket, two standard identities are useful:

$$\frac{\partial\dot{\vec{r}}_j}{\partial\dot{q}_i} = \frac{\partial\vec{r}_j}{\partial q_i}, \qquad \frac{d}{dt}\left(\frac{\partial\vec{r}_j}{\partial q_i}\right) = \frac{\partial\dot{\vec{r}}_j}{\partial q_i}. \tag{10}$$

Using these identities, together with the kinetic energy

$$T = \frac{1}{2}\sum_{j=1}^{N}m_j\,\dot{\vec{r}}_j\cdot\dot{\vec{r}}_j, \tag{11}$$

the inertial term becomes

$$\sum_{j=1}^{N}m_j\ddot{\vec{r}}_j\cdot\frac{\partial\vec{r}_j}{\partial q_i} = \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{q}_i}\right) - \frac{\partial T}{\partial q_i}. \tag{12}$$

Therefore, d'Alembert's principle takes the form

$$\sum_{i=1}^{n} \left[Q_i - \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) + \frac{\partial T}{\partial q_i} \right] \delta q_i = 0, \tag{13}$$

which must hold for every admissible virtual displacement $(\delta q_1, \dots, \delta q_n)$. This is the starting point for everything that follows.

1.2 A Careful Statement of the Principle

Define the quantities

$$A_i = Q_i - \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) + \frac{\partial T}{\partial q_i}. \tag{14}$$

Then d'Alembert's principle states that

$$\sum_{i=1}^{n} A_i \, \delta q_i = 0, \tag{15}$$

or, in vector notation,

$$A \cdot \delta q = 0, \tag{16}$$

for every **admissible virtual displacement** δq.

The important point is this:

> D'Alembert's principle is not a statement about arbitrary variations. It is a statement about all virtual displacements that are **compatible** with the constraints.

If the coordinates q_i are independent, then the virtual displacements δq_i are arbitrary and independent, and one may conclude that $A_i = 0$ for each i.

If constraints are present, however, the virtual displacements are restricted. In that case, from $A \cdot \delta q = 0$ one cannot directly conclude that every A_i is zero; one can only conclude that A is orthogonal to every admissible virtual displacement.

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2. Two Different Uses of "Generalized Coordinates"

A common source of confusion is that the phrase **generalized coordinates** is used in two different ways.

2.1 Case A: Independent Generalized Coordinates

Suppose the system has n degrees of freedom, and we choose $q_1, \dots, q_n$ as **independent generalized coordinates**. Then the corresponding virtual displacements $\delta q_1, \dots, \delta q_n$ are also independent. In this case, d'Alembert's principle leads to

$$\sum_{i=1}^{n}\left[Q_i-\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right)+\frac{\partial T}{\partial q_i}\right]\delta q_i=0. \tag{17}$$

Since all δq_i are independent, each coefficient must vanish:

$$Q_i-\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right)+\frac{\partial T}{\partial q_i}=0. \tag{18}$$

Equivalently,

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right)-\frac{\partial T}{\partial q_i}=Q_i, \qquad i=1,\dots,n. \tag{19}$$

This is the usual form of Lagrange's equations.

2.2 Case B: Redundant Generalized Coordinates

Sometimes, however, we deliberately use more coordinates than the number of degrees of freedom. Suppose we describe the system using $q_1,\dots,q_n$, but these coordinates are subject to m **holonomic constraints (完整约束)**:

$$f_\alpha(q,t)=0, \qquad \alpha=1,\dots,m. \tag{20}$$

Then the actual number of degrees of freedom is $n-m$, assuming the constraints are independent. In this situation, the coordinates q_i are **not all independent**.

A virtual displacement must be compatible with the constraints. Since a virtual displacement is taken at fixed time, we have

$$\delta f_\alpha=\sum_{i=1}^{n}\frac{\partial f_\alpha}{\partial q_i}\,\delta q_i=0, \qquad \alpha=1,\dots,m. \tag{21}$$

Thus the virtual displacements δq_i are not independent. This is precisely the situation in which **Lagrange multipliers (拉格朗日乘子)** are needed.

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3. First Explanation: Independent Coordinates Give the Usual Lagrange Equations

If we choose coordinates adapted to the constraints, then the coordinates are independent. For example, suppose the system has r degrees of freedom, and we choose independent coordinates $s_1,\dots,s_r$. Then the virtual displacements $\delta s_1,\dots,\delta s_r$ are arbitrary and independent. Applying d'Alembert's principle in these independent coordinates gives

$$\sum_{a=1}^{r}\left[Q_a-\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{s}_a}\right)+\frac{\partial T}{\partial s_a}\right]\delta s_a=0. \tag{22}$$

Because the δs_a are independent, the coefficients must vanish:

$$Q_a - \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{s}_a}\right) + \frac{\partial T}{\partial s_a} = 0. \tag{23}$$

Therefore,

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{s}_a}\right) - \frac{\partial T}{\partial s_a} = Q_a, \qquad a = 1, \dots, r. \tag{24}$$

These are the ordinary Lagrange equations in independent generalized coordinates.

The advantage of this approach is that the equations contain no constraint forces explicitly, because the constraints have already been built into the choice of coordinates. The disadvantage is that finding such independent coordinates may be difficult.

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4. Second Explanation: Redundant Coordinates Lead to Lagrange Multipliers

Now suppose instead that we describe the same system using redundant coordinates

$$\vec{q} = (q_1, \dots, q_n), \tag{25}$$

together with m independent holonomic constraints

$$f_\alpha(\vec{q}, t) = 0, \quad \alpha = 1, \dots, m. \tag{26}$$

The corresponding virtual displacement in generalized-coordinate space is

$$\delta\vec{q} = (\delta q_1, \dots, \delta q_n). \tag{27}$$

Since a virtual displacement is taken at fixed time, the admissible virtual displacements satisfy

$$\delta f_\alpha = 0, \tag{28}$$

that is,

$$\sum_{i=1}^{n} \frac{\partial f_\alpha}{\partial q_i}\, \delta q_i = 0, \quad \alpha = 1, \dots, m. \tag{29}$$

Equivalently, in vector form,

$$\nabla f_\alpha \cdot \delta\vec{q} = 0, \tag{30}$$

where

$$\nabla f_\alpha = \left(\frac{\partial f_\alpha}{\partial q_1}, \dots, \frac{\partial f_\alpha}{\partial q_n}\right). \tag{31}$$

Thus $\delta\vec{q}$ must lie in the tangent space to the constraint surface.

D'Alembert's principle says that

$$\vec{A} \cdot \delta\vec{q} = 0 \tag{32}$$

for every such admissible virtual displacement $\delta\vec{q}$, where

$$\vec{A} = (A_1, \dots, A_n). \tag{33}$$

Therefore, $\vec{A}$ **is orthogonal to the tangent space of the constraint surface**.

The normal space to the constraint surface is spanned by the gradients of the constraint functions,

$$\nabla f_1, \dots, \nabla f_m. \tag{34}$$

Assuming the constraints are independent, these gradient vectors are linearly independent.

> **Note.** The gradients of the constraints give the directions pointing "out of" the allowed surface. Combining these gradient directions gives all possible normal directions to the surface. In the case of independent constraints, the gradient vectors of the constraint functions are linearly independent and form a **basis for the normal space to the constraint surface**. However, this basis is generally not orthonormal.
>
> 约束函数的梯度指向"离开"约束曲面的方向；把这些梯度方向组合起来，就可以得到所有可能的法向方向。在独立约束的情况下，约束函数的梯度向量线性无关，并构成约束曲面法空间的一组基矢；但这组基矢一般不是正交归一的。

As $\vec{A}$ must lie in the normal space, there exist multipliers λ_α such that

$$\vec{A} = \sum_{\alpha=1}^{m} \lambda_\alpha \nabla f_\alpha. \tag{35}$$

Using component notation, this means

$$(A_1, \dots, A_n) = \sum_{\alpha=1}^{m} \lambda_\alpha \left(\frac{\partial f_\alpha}{\partial q_1}, \dots, \frac{\partial f_\alpha}{\partial q_n} \right). \tag{36}$$

Equivalently, for each component,

$$A_i = \sum_{\alpha=1}^{m} \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}, \qquad i = 1, \dots, n. \tag{37}$$

Substituting the definition of A_i, we get

$$Q_i - \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) + \frac{\partial T}{\partial q_i} = \sum_{\alpha=1}^{m} \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}. \tag{38}$$

Rearranging,

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = Q_i - \sum_{\alpha=1}^{m} \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}. \tag{39}$$

If we redefine the multipliers by changing their sign,

$$\lambda_\alpha \mapsto -\lambda_\alpha, \tag{40}$$

then the same equations can be written as

$$\boxed{\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = Q_i + \sum_{\alpha=1}^{m} \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}} \tag{41}$$

Here Q_i is the applied generalized force, while the sum over λ_α collects the generalized constraint forces.

Thus the sign of the multiplier term is a matter of convention.

Together with the constraint equations

$$f_\alpha(q,t) = 0, \qquad \alpha = 1, \dots, m, \tag{42}$$

these form a closed system of equations for $q_i(t)$ and $\lambda_\alpha(t)$.

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5. Summary

The key distinction is the following.

- If the generalized coordinates are **independent**, then the virtual displacements δq_i are independent. Therefore, from d'Alembert's principle one obtains

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = Q_i. \tag{43}$$

- If the generalized coordinates are **redundant**, then the virtual displacements δq_i are constrained. Therefore, one cannot set each coefficient equal to zero. Instead, the coefficient vector must lie in the normal space to the constraint surface, giving

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = Q_i + \sum_{\alpha=1}^{m} \lambda_\alpha \frac{\partial f_\alpha}{\partial q_i}, \tag{44}$$

  up to the sign convention for λ_α.

In short:

> Independent coordinates eliminate the constraints by construction. Redundant coordinates keep the constraints explicit and introduce Lagrange multipliers to account for the missing independence of the virtual displacements.

5.1 Supplementary Remarks in Chinese

达朗贝尔原理（d'Alembert's principle）本质上可以看作一种**虚功原理**。它说的是：在系统的真实运动过程中，外加力与惯性力之差，对任意允许的虚位移所作的总虚功为零。这里最重要的一点是，**虚位移必须与约束相容**，也就是说，虚位移必须满足系统约束所允许的条件；但是，**达朗贝尔原理本身并不要求这些虚位移的各个分量 δq_i 一定彼此独立**。

因此，关键问题在于我们如何选择广义坐标。大体上可以分为两种情况。

第一种情况是：我们已经利用约束条件选取了一组**独立的广义坐标**。例如，对于一个受完整约束（holonomic constraints）限制的系统，如果约束方程已经被用来消去多余变量，那么剩下的广义坐标数目就等于系统的自由度数。在这种情况下，$\delta q_1, \dots, \delta q_n$ 是彼此独立的。于是，如果

$$\sum_{i=1}^{n} A_i \delta q_i = 0 \tag{45}$$

对任意独立的 δq_i 都成立，那么就可以推出

$$A_i = 0, \qquad i = 1, \dots, n. \tag{46}$$

这正是通常从达朗贝尔原理推出拉格朗日方程（Lagrange equations）时所采用的情形。

第二种情况是：我们仍然使用一组**冗余坐标**，或者某些约束条件尚未被完全消去。此时虽然仍然可以写出

$$A \cdot \delta q = 0, \tag{47}$$

但是 δq_i 不再是任意且彼此独立的变量，而必须满足由约束条件带来的关系。例如，某些允许的虚位移可能满足

$$\sum_i c_i \delta q_i = 0. \tag{48}$$

在这种情况下，不能直接推出每一个 A_i 都等于零。我们只能说，**向量 A 与所有允许的虚位移方向正交**。若要进一步得到运动方程，通常需要引入拉格朗日乘子（Lagrange multipliers），把约束条件或约束力的影响显式地写入方程中。

关于“广义坐标”（generalized coordinates）这个术语，不同教材的侧重点确实略有不同。在许多经典力学教材中，尤其是在处理完整约束并推导拉格朗日方程时，广义坐标通常被选为一组**独立坐标**。也就是说，人们常常先利用约束方程消去多余的笛卡尔坐标（Cartesian coordinates），最后保留与系统自由度数目相同的一组变量。在这种标准语境下，q_i 往往默认就是独立的广义坐标。

不过，从更宽泛的意义上说，“广义坐标”这个词也强调这些坐标不必是普通的笛卡尔坐标，而可以是任何方便描述系统构型的变量，例如角度、弧长、欧拉角等。因此，有些教材在引入广义坐标时，更强调“广义”这一点，也就是它们可以是非笛卡尔坐标；而另一些教材则会同时强调，为了唯一描述一个具有 n 个自由度的系统，通常应当选取 n 个彼此独立的广义坐标。

所以，更准确地说：在推导标准形式的拉格朗日方程时，广义坐标通常已经被选为与自由度数目相同的独立坐标；但从术语本身来看，“广义坐标”主要强调它们不必是笛卡尔坐标，而可以是任何适合描述系统构型的变量。如果使用的是冗余坐标，那么它们也可以在较宽泛的意义下被称为广义坐标，但这时其虚位移 δq_i 就不能被当作独立变量处理。因此，达朗贝尔原理的核心是“δq 是否是与约束相容的允许虚位移”。

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6. Particle Sliding on a Hemisphere

A particle of mass m is released on the outside of a smooth fixed hemisphere of radius R, from a point very close to but not exactly at the top. The initial displacement is enough to make the particle slide, and we may assume its initial velocity is zero. We seek the angle θ_0, measured from the vertical, at which the particle loses contact with the surface.

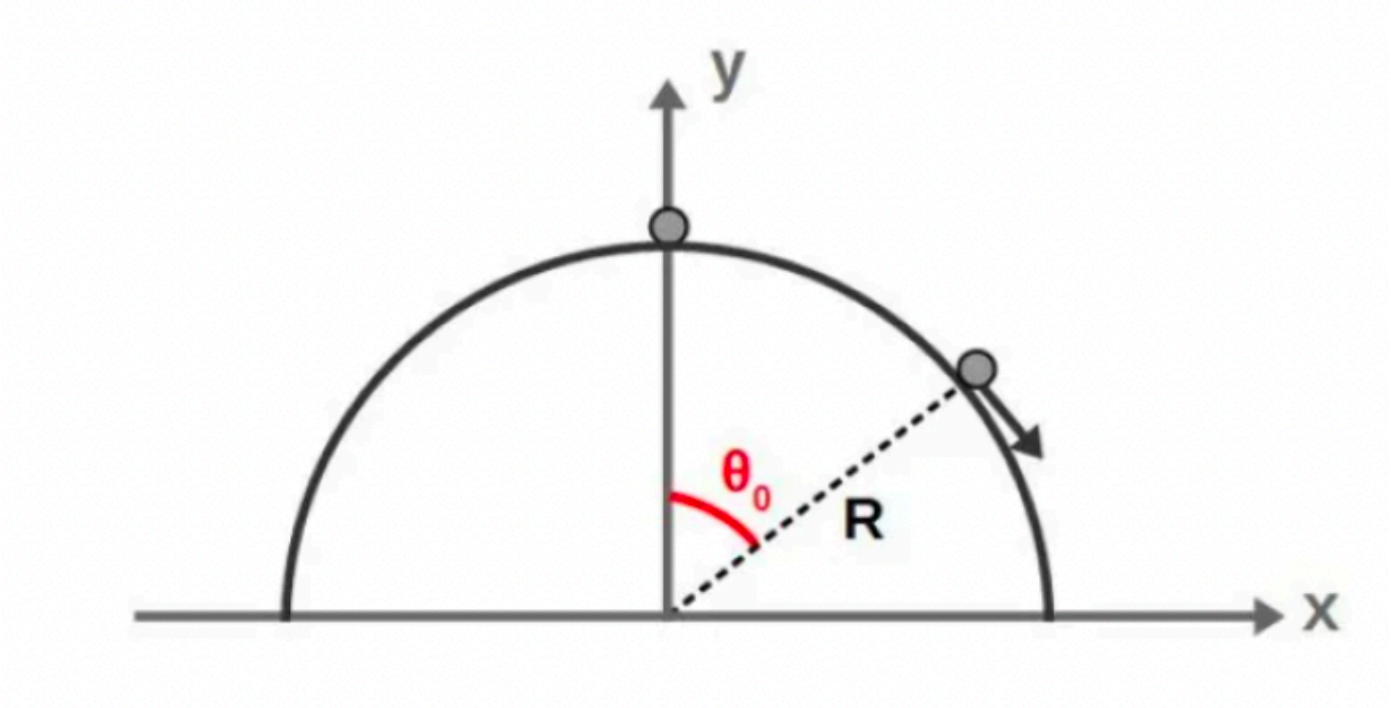


Figure 1. A particle slides on the outside of a smooth hemisphere and loses contact with it where the normal force vanishes.

The Lagrangian in polar coordinates (r, θ), with θ measured from the vertical, is

$$L = \frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right) - mgr \cos\theta. \tag{49}$$

The holonomic constraint is

$$g(r) = r - R = 0. \tag{50}$$

The constrained Lagrange equations are

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = \lambda \frac{\partial g}{\partial r} = \lambda, \tag{51}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = \lambda \frac{\partial g}{\partial \theta} = 0. \tag{52}$$

Evaluating the derivatives,

$$m\ddot{r} - mr\dot{\theta}^2 + mg\cos\theta = \lambda, \tag{53}$$

$$2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta} - mgr\sin\theta = 0. \tag{54}$$

We now impose the constraint $r = R$, which implies $\dot{r} = \ddot{r} = 0$. The radial equation becomes

$$-mR\dot{\theta}^2 + mg\cos\theta = \lambda, \tag{55}$$

so

$$\dot{\theta}^2 = \frac{g\cos\theta}{R} - \frac{\lambda}{mR}. \tag{56}$$

The angular equation reduces to

$$mR^2\ddot{\theta} - mgR\sin\theta = 0 \quad \Longrightarrow \quad \ddot{\theta} = \frac{g\sin\theta}{R}. \tag{57}$$

> **Physical meaning of λ.** The Lagrange multiplier λ is the magnitude of the normal force that the hemisphere exerts on the particle. When λ drops to zero, the constraint is about to break; the particle can no longer press against the surface and therefore leaves the sphere.

The particle leaves the surface precisely when the normal force vanishes:

$$\lambda = 0. \tag{58}$$

Hence, at the instant of departure,

$$\dot{\theta}^2 = \frac{g\cos\theta_0}{R}. \tag{59}$$

To relate θ_0 to the initial condition, we differentiate the radial expression for $\dot{\theta}^2$ with respect to time:

$$\frac{d}{dt}\dot{\theta}^2 = 2\dot{\theta}\ddot{\theta}. \tag{60}$$

Using $\dot{\theta}^2 = (g\cos\theta)/R - \lambda/(mR)$,

$$2\dot{\theta}\ddot{\theta} = -\frac{g}{R}\sin\theta\,\dot{\theta} - \frac{1}{mR}\frac{d\lambda}{dt}. \tag{61}$$

Since $\ddot{\theta} = g\sin\theta/R$, the left-hand side becomes

$$2\dot{\theta}\left(\frac{g\sin\theta}{R}\right) = -\frac{g}{R}\sin\theta\,\dot{\theta} - \frac{1}{mR}\frac{d\lambda}{dt}, \tag{62}$$

or, after rearranging,

$$\frac{3g}{R}\sin\theta\,\dot{\theta} = -\frac{1}{mR}\frac{d\lambda}{dt}. \tag{63}$$

Multiplying by dt and using $d\theta = \dot{\theta}\,dt$,

$$3g\sin\theta\,d\theta = -\frac{1}{m}\,d\lambda. \tag{64}$$

Integrating from the initial position $(\theta = 0,\ \lambda_0)$ to the leaving point $(\theta_0,\ 0)$,

$$3g\int_0^{\theta_0}\sin\theta\,d\theta = -\frac{1}{m}\int_{\lambda_0}^{0} d\lambda. \tag{65}$$

This gives

$$3g\big(1-\cos\theta_0\big)=\frac{\lambda_0}{m}. \tag{66}$$

At the top of the sphere, the particle is initially at rest ($\dot\theta=0$ when $\theta=0$), so the radial equation yields $\lambda_0 = mg$. Therefore

$$3g\big(1-\cos\theta_0\big)=g, \tag{67}$$

which simplifies to

$$\boxed{\cos\theta_0=\frac{2}{3}} \tag{68}$$

or

$$\theta_0\approx 48.2^\circ. \tag{69}$$

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7. Particle Constrained to Move on a Spiral

A particle of mass m moves in a horizontal plane on a spiral whose equation in polar coordinates is

$$r=a\theta, \tag{70}$$

where a is a positive constant. There is no external potential, so apart from its inertia the only force acting on the particle is the force of constraint. We seek the equation of motion and the constraint force that keeps the particle on the spiral.

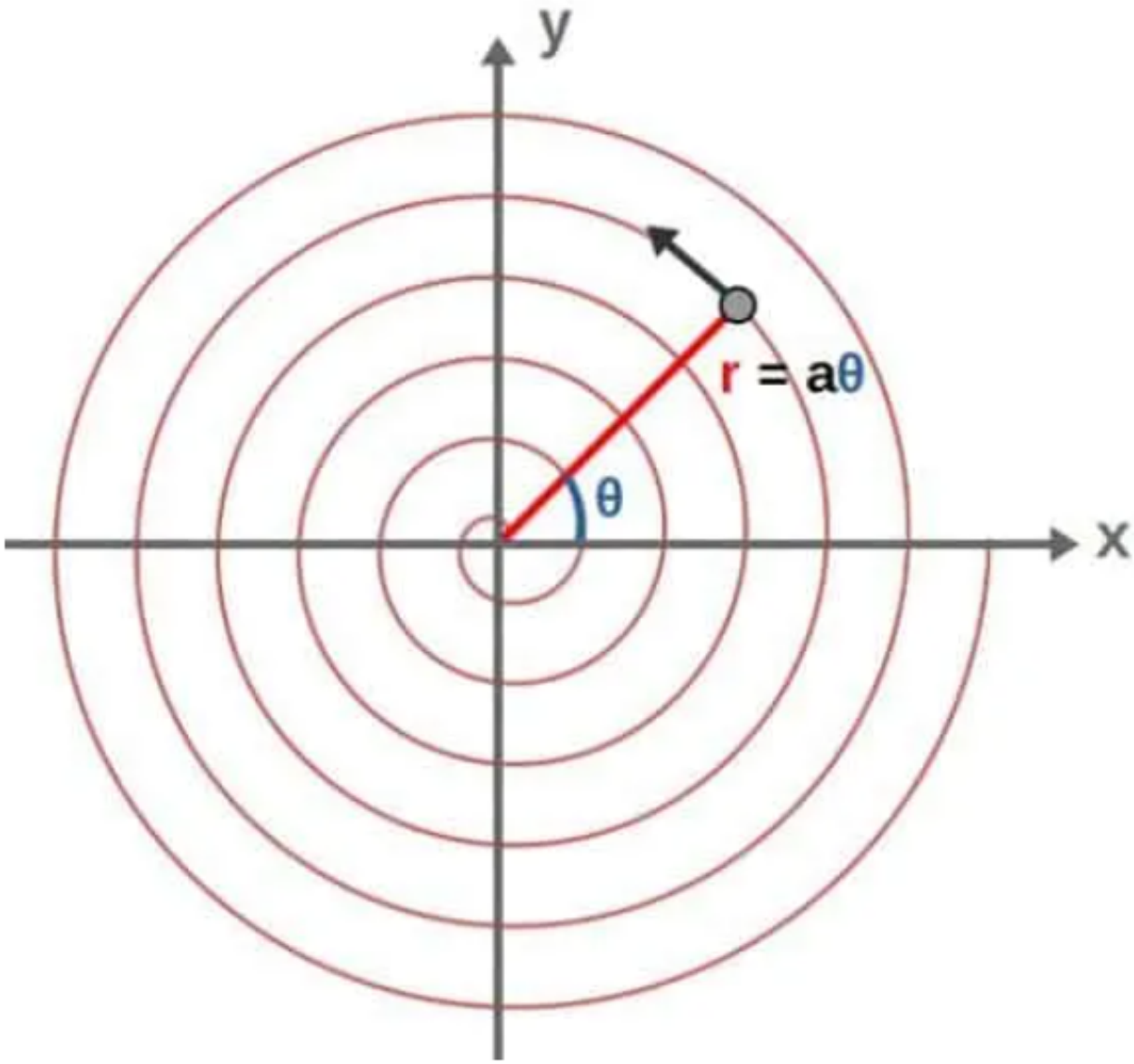


Figure 2. A particle constrained to move along the Archimedean spiral $r = a\theta$.

The unconstrained Lagrangian consists of the kinetic energy alone, which in polar coordinates is

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right). \tag{71}$$

At this stage the constraint has not been used, so this is the Lagrangian of a freely moving particle in polar coordinates; it knows nothing about the spiral.

The holonomic constraint is

$$g(r, \theta) = r - a\theta = 0. \tag{72}$$

If we only wanted the equation of motion, we could substitute $r = a\theta$ into L and obtain a single Lagrange equation for θ. That would, however, eliminate the constraint force. To keep track of it, we retain both coordinates and introduce a Lagrange multiplier.

Since there is only one constraint, the constrained Lagrange equations read

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = \lambda\frac{\partial g}{\partial r} = \lambda, \tag{73}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = \lambda \frac{\partial g}{\partial \theta} = -a\lambda. \tag{74}$$

Evaluating the derivatives,

$$m\ddot{r} - mr\dot{\theta}^2 = \lambda, \tag{75}$$

$$2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta} = -a\lambda. \tag{76}$$

Therefore the two unconstrained equations of motion are

$$\ddot{r} = \frac{\lambda}{m} + r\dot{\theta}^2, \tag{77}$$

$$\ddot{\theta} = -\frac{a\lambda}{mr^2} - \frac{2\dot{r}\dot{\theta}}{r}. \tag{78}$$

We now impose the constraint $r = a\theta$, which implies

$$\dot{r} = a\dot{\theta}, \qquad \ddot{r} = a\ddot{\theta}. \tag{79}$$

The two equations become

$$a\ddot{\theta} = \frac{\lambda}{m} + a\theta\dot{\theta}^2 \quad \Longrightarrow \quad \ddot{\theta} = \frac{\lambda}{ma} + \theta\dot{\theta}^2, \tag{80}$$

$$\ddot{\theta} = -\frac{a\lambda}{ma^2\theta^2} - \frac{2(a\dot{\theta})\dot{\theta}}{a\theta} \quad \Longrightarrow \quad \ddot{\theta} = -\frac{\lambda}{ma\theta^2} - \frac{2\dot{\theta}^2}{\theta}. \tag{81}$$

Both expressions are equations for $\ddot{\theta}$, so they must be equal:

$$\frac{\lambda}{ma} + \theta\dot{\theta}^2 = -\frac{\lambda}{ma\theta^2} - \frac{2\dot{\theta}^2}{\theta}. \tag{82}$$

Solving for the multiplier,

$$\lambda\left(1 + \frac{1}{\theta^2}\right) = -ma\dot{\theta}^2\left(\theta + \frac{2}{\theta}\right), \tag{83}$$

and hence

$$\lambda = -\frac{2\theta + \theta^3}{1 + \theta^2}\, ma\dot{\theta}^2. \tag{84}$$

Substituting this back into either equation of motion gives

$$\ddot{\theta} = -\frac{\theta}{1 + \theta^2}\, \dot{\theta}^2. \tag{85}$$

The equation of motion and the Lagrange multiplier are therefore

$$\boxed{\ddot{\theta} = -\frac{\theta}{1+\theta^2}\,\dot{\theta}^2} \tag{86}$$

and

$$\lambda = -\frac{2\theta + \theta^3}{1+\theta^2}\, ma\dot{\theta}^2. \tag{87}$$

> **Physical meaning of λ.** The generalized constraint force is $\lambda \nabla g$, so its r- and θ-components are

$$Q_r^{(c)} = \lambda \frac{\partial g}{\partial r} = \lambda = -\frac{2\theta + \theta^3}{1+\theta^2}\, ma\dot{\theta}^2, \tag{88}$$

$$Q_\theta^{(c)} = \lambda \frac{\partial g}{\partial \theta} = -a\lambda = \frac{2\theta + \theta^3}{1+\theta^2}\, ma^2\dot{\theta}^2. \tag{89}$$

These are the components of the force that the spiral must exert on the particle in order to keep it on the curve. For $\theta > 0$ and $\dot{\theta} \neq 0$, the radial component is negative, so the spiral pushes the particle inward, while the azimuthal component is positive.

The equation of motion for θ has no useful closed-form solution for $\theta(t)$. It can, however, be integrated once. Writing $\ddot{\theta} = \dot{\theta}\, d\dot{\theta}/d\theta$,

$$\dot{\theta}\frac{d\dot{\theta}}{d\theta} = -\frac{\theta}{1+\theta^2}\,\dot{\theta}^2, \tag{90}$$

so that

$$\frac{d\dot{\theta}}{\dot{\theta}} = -\frac{\theta\, d\theta}{1+\theta^2}. \tag{91}$$

Integrating,

$$\ln\dot{\theta} = -\frac{1}{2}\ln\left(1+\theta^2\right) + \text{const}, \tag{92}$$

or

$$\dot{\theta} = \frac{C}{\sqrt{1+\theta^2}}. \tag{93}$$

This is consistent with energy conservation: the constraint is ideal, so the constraint force does no work, and the speed

$$v^2 = \dot{r}^2 + r^2\dot{\theta}^2 = a^2\dot{\theta}^2\left(1+\theta^2\right) = a^2C^2 \tag{94}$$

is constant, as it must be for a free particle. Using $d\theta = \dot{\theta}\, dt$, we obtain

$$dt = \frac{\sqrt{1+\theta^2}}{C}\, d\theta, \tag{95}$$

which integrates to

$$\boxed{\theta\sqrt{1+\theta^2} + \operatorname{arcsinh}\theta = At + B} \tag{96}$$

for constants A and B. This is a relation between t and θ; it cannot be inverted to give $\theta(t)$ in closed form. As is usually the case in Lagrangian mechanics, the equations of motion themselves, together with the expression for the constraint force, are the real result.